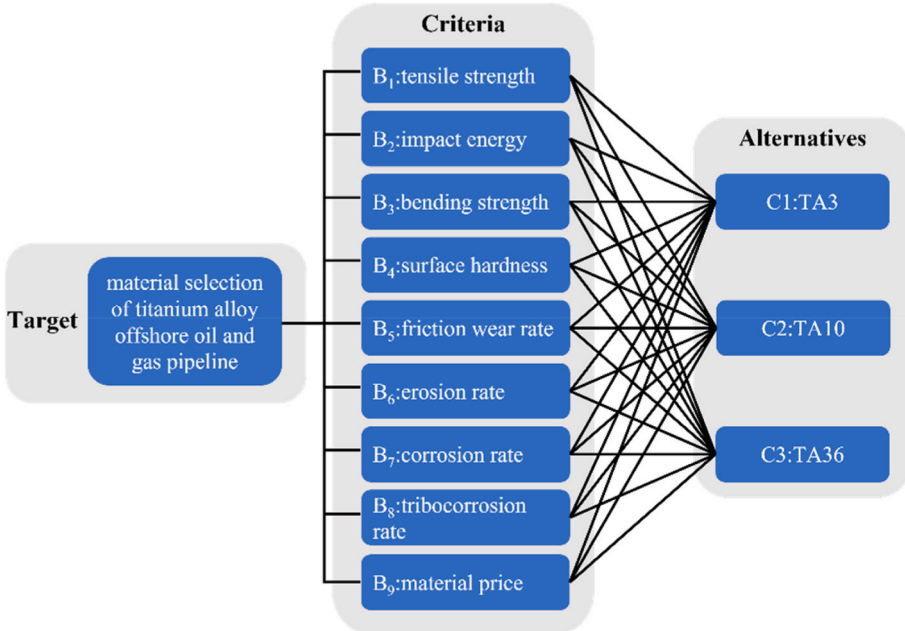


**Fig. 9.** Hierarchical structure model.

**Table 7**
Relative importances of factors [73].

| Descriptions ($i$ over $j$) | Relative importances ($a_{ij}$) |
|---|---|
| Equal importance | 1 |
| Moderate importance | 3 |
| Strong importance | 5 |
| Very strong importance | 7 |
| Absolute importance | 9 |
| Intermediate values | 2,4,6,8 |

$$wi = \frac{\left(\prod_{j=1}^{n} a_{ij}\right)^{1/n}}{\sum_{i=1}^{n}\left(\prod_{j=1}^{n} a_{ij}\right)^{1/n}} \tag{4}$$

where $wi$ indicates the weight of item I; $n$ indicates the order of matrix; $i, j = 1, 2, \ldots n$.

Step 3: Weight vector calculation and consistency test of the criterion layer. The consistency of the importance of each criterion is tested. In order to ensure that the evaluation of the pairwise comparison matrix is reasonable and acceptable, a consistency check is performed. Let $C$ represent an $n$-dimensional column vector that describes the sum of the weighted values of the criterion importance, then we get Eq. (5) [75]:

$$c = [ci] = A.W^{T}_{n\times 1} \tag{5}$$

where.

$$A.W^{T} = \begin{bmatrix} 1 & a_{12} & \cdots & a_{1n} \\ a_{21} & 1 & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & 1 \end{bmatrix} \bullet [w_1, w_2, \cdots, w_n]^{T} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \tag{6}$$

The consistency value of the criterion cluster can be expressed as $CV = [cv]_{n\times 1}$, where the typical elements $cv_i$ are defined as Eq. (7) [76]:

$$CV_i = \frac{c_i}{w_i} \tag{7}$$

However, in order to avoid the inconsistencies among different measurement scales in the evaluation process, Saaty recommended to use the maximum eigenvalue ($l_{max}$) to evaluate the effectiveness of the measurement. The maximum eigenvalue can be determined as Eq. (8) [76]:

$$l_{max} = \frac{\sum_{i=1}^{n} cv_i}{n} \tag{8}$$

Based on the largest eigenvalue, the consistency index (CI) is determined as Eq. (9) [52]:

$$CI = \frac{l_{max} - n}{n - 1} \tag{9}$$

where if $CI = 0$, the evaluation of the pairwise comparison matrix is consistent. It is worth noting that the closer the maximum eigenvalue is to $n$, the more consistent the evaluation is. In general, the consistency ratio ($CR$) can be used as a guide for consistency check Eq. (10) [75].

$$CR = \frac{CI}{RI} \tag{10}$$

where $RI$ is the average random index of the values obtained by comparing the values of different orders of matrices in pairs (Eq. (11)). If the $CR$ value is less than 0.1, it is considered that the importance evaluation of the index is reasonable [77]. When the final consistency ratio exceeds 0.1, the evaluation method must be iterated to achieve the effective consistency. The measurement of consistency can be used to evaluate the consistency of decision makers and the consistency of the whole level.

$$RI = 1.98\frac{n-2}{n} \tag{11}$$

Step 4: The alternative layer judgment matrix Bi is established according to the same method of the criterion layer judgment matrix. The weight $Wj$ of the alternative under a single criterion is calculated and then a consistency test is performed.

Step 5: Synthetic weight and overall ranking. In terms of the criteria and targets of the selection, the output of each alternative is generally expressed as Eq. (12) [78]: